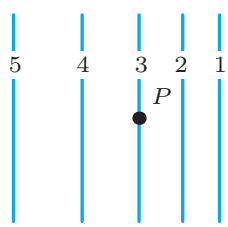
Suggested problems: 1-11 odd, 41, 43

Problem 1: If $f(x, y) = x^4y^2 - x^3y$, find:

(a) $f_x(x, y) = 4x^3y^2 - 3x^2y$
(b) $f_y(x, y) = 2x^4y - x^3$
(c) $f_{xx}(x, y) = 12x^2y^2 - 6xy$
(d) $f_{xy}(x, y) = 8x^3y - 3x^2$
(e) $f_{yx}(x, y) = 8x^3y - 3x^2$
(f) $f_{yy}(x, y) = 2x^4$

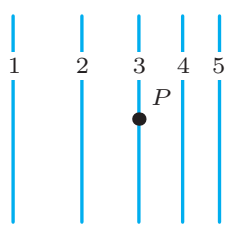
Problem 2: For each of the contour plots below, decide if the signs of the various partial derivatives at the point P are positive or negative, or if the derivatives are zero. The derivatives are $f_x(P)$, $f_y(P)$, $f_{xx}(P)$, $f_{yy}(P)$, $f_{xy}(P)$.

(a)



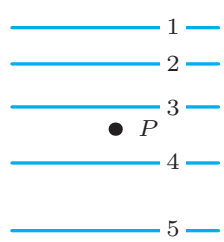
$f_x(P) < 0,\ f_y(P) = 0,\ f_{xx}(P) < 0,\ f_{yy}(P) = 0,\ f_{xy}(P) = 0$

(b)



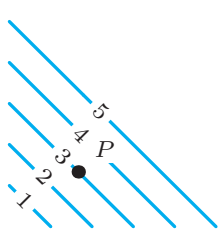
$f_x(P) > 0,\ f_y(P) = 0,\ f_{xx}(P) > 0,\ f_{yy}(P) = 0,\ f_{xy}(P) = 0$

(c)



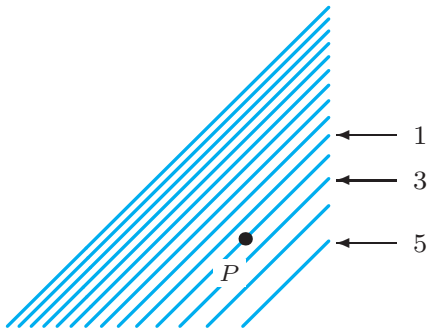
$f_x(P) = 0,\ f_y(P) < 0,\ f_{xx}(P) = 0,\ f_{yy}(P) < 0,\ f_{xy}(P) = 0$

(d)



$f_x(P) > 0,\ f_y(P) > 0,\ f_{xx}(P) = 0,\ f_{yy}(P) = 0,\ f_{xy}(P) = 0$

(e)



$f_x(P) > 0,\ f_y(P) < 0,\ f_{xx}(P) < 0,\ f_{yy}(P) < 0,\ f_{xy}(P) > 0$